PHYSICAL ACTIVITY



Poor functional capacity and physical fitness are associated with poor surgical outcomes including prolonged hospital length of stay and increased risk of postoperative complications. Increasing physical fitness can improve resilience and recovery after surgery. [1,2]

Screening Tools

The Physical Activity Vital Sign (PAVS) Calculator is a quick and easy way to flag sedentary patients for referral and counseling.

The Physical Activity Vital Sign Calculator

1. On average, how many days per week do you engage in moderate to strenuous exercise (like a brisk walk)?	_____ days
2. On average, how many minutes do you engage in exercise at this level?	_____ minutes
Total minutes per week of physical activity (multiply #1 by #2)	_____ minutes *per week*

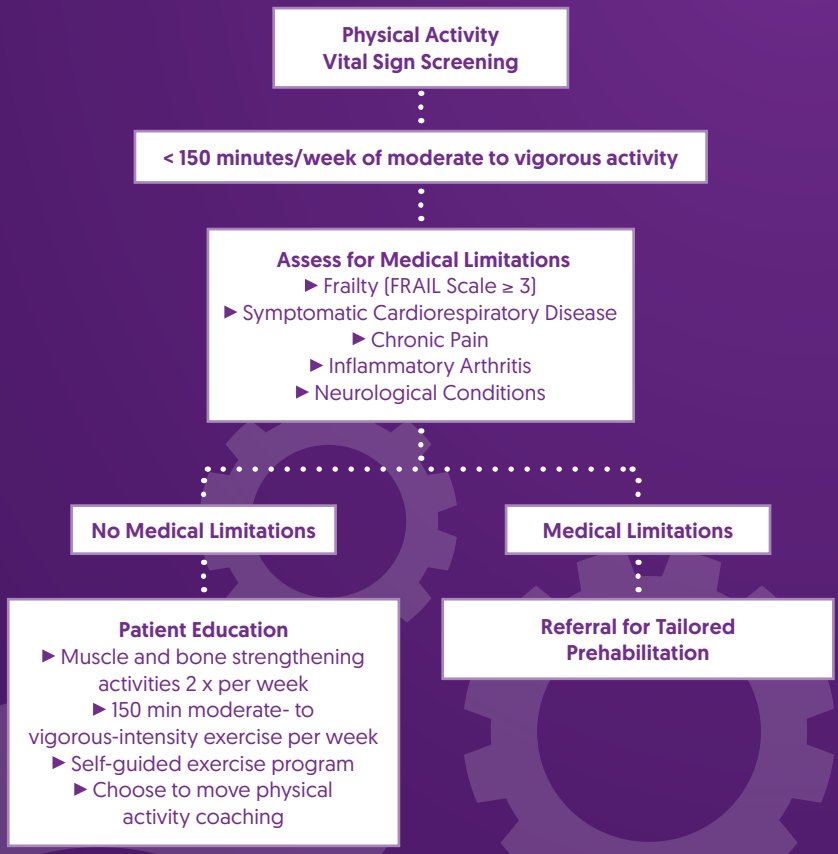
Using the Physical Activity Vital Sign

- National guidelines recommend 150 minutes per week of moderate intensity physical activity. Moderate intensity activity is usually done where an individual can talk, but cannot "sing". Examples include: brisk walking, slow biking, general gardening, and ballroom dancing.
- In place of moderate intensity activity, an individual can also complete 75 minutes of vigorous intensity physical activity. Vigorous intensity activity is done at a pace where individuals can no longer talk and are somewhat out of breath. Examples include: swimming laps, playing singles tennis, and fast bicycling.
- Individuals can also achieve 150 "minutes" through a combination of moderate and vigorous intensity physical activity, with 1 minute of vigorous activity being equal to 2 minutes of moderate activity.
- If activity is done throughout the day, individuals are encouraged to perform activity in "bouts" that are at least 10 minutes in length.
- If your patient is NOT achieving 150 minutes a week of physical activity, advise them to gradually increase either their frequency or duration until they are capable of safely performing 10 minutes bouts of activity and achieve national recommendations.

The Physical Activity Vital Sign – Other Considerations

- A comprehensive assessment of physical activity should include promotion of active living throughout the day to reduce sedentary time, as well as muscle strengthening and flexibility exercises as recommended by the Physical Activity Guidelines for Americans.
- If you wish to add a question on muscle strengthening activities, we would recommend the following question: **How many days a week do you perform muscle strengthening exercises, such as bodyweight exercises or resistance training?** _____ days

PHYSICAL ACTIVITY ALGORITHM



Patient Education

- Emphasize the importance of muscle- and bone-strengthening activities at least 2 days per week to increase strength and functional capacity prior to surgery (e.g., STABL Exercises) [5]
- Recommend at least 150 min of moderate- to vigorous-intensity aerobic physical activity per week, in bouts of 10 minutes or more
  - Moderate intensity activity: brisk walk or riding a bicycle; you can talk but not sing
  - Vigorous intensity activity: jogging or swimming; you are out of breath and can talk but do not want to
- Refer to the FREE Choose to Move program for personalized physical activity coaching and support. Participants set goals, build a tailored physical activity plan, and connect with peers while receiving information on health and wellness. In-person and online programs available.

Referral for Tailored Prehabilitation

- Consider referral to physiotherapy for tailored preoperative exercise guidance for patients with medical limitations such as:
  - Frailty *(See frailty)*
  - Symptomatic cardiorespiratory disease
  - Chronic pain
  - Inflammatory arthritis
  - Neurological conditions
- Refer to online provider resources for physiotherapy resources, referral templates, and information on who can access physio for free in BC